

15

Body Composition and Nutrition for Sport

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In Closing

Defensive linemen in the National Football League are large men, especially the defensive tackles who play the so-called 0-technique position in the middle of the line. By nature of their role, these behemoths must be hard to push off the line of scrimmage, yet agile enough to fill running lanes and rush the opposing quarterback. Dontari Poe is one of the best at his position and was rewarded by signing a 1-year \$8 million contract with the Atlanta Falcons in 2017. However, there was a catch. Five hundred thousand dollars of that amount was tied to a weight clause that could be earned only if he reportedly weighed in at 149.7 kg (330 lb), a weight that may optimize his athletic performance. During the 2016 season, Poe was listed by the Chiefs as weighing 157.3 kg (346 lb), so he had some work to do to reach his goal. Similarly, a weight clause was included in his \$27 million deal in 2018 with the Carolina Panthers, which included quarterly bonuses for meeting successful weight targets. Attesting to his overall athleticism, although he is a defensive player, Poe holds the records for being the heaviest NFL player to run for a touchdown and the heaviest player to throw a pass for a touchdown! Many athletes compete in sports for which management of weight—and body composition—is critical, including wrestling, boxing, gymnastics, and distance running. Yet extreme changes may be detrimental to an athlete's performance and overall health. Achieving optimal body weight and composition is a critical component of successful sport performance.

Coaches and athletes today are acutely aware of the importance of achieving and maintaining optimal body weight for peak performance in sport. Appropriate size, physique, and body composition are critical to success in almost all athletic endeavors. Compare the specific performance requirements of the 152 cm, 45 kg (5 ft, 100 lb) Olympic gymnast and those of the 206 cm, 147 kg (6 ft 9 in., 325 lb) defensive lineman in professional football. Although size and body build can be altered only slightly, body composition can change substantially with dieting and exercise. Resistance training can substantially increase muscle mass, and sound dieting combined with vigorous exercise can significantly decrease body fat. Such changes can be of major importance in achieving optimal athletic performance.

Peak performance also requires a careful dietary balance of the essential nutrients. The U.S. government has established standards for optimal nutrient intake that are termed Dietary Reference Intakes

(DRIs). The DRIs provide estimates of the range of intakes of various food substances needed to maintain good health. However, these guidelines were not written for extremely active people or athletes, and the nutrition needs of athletes can exceed the DRIs considerably. Individual caloric needs are extremely variable, depending on the athlete's size, sex, training volume, and sport. Cyclists competing in the Tour de France and Norwegian cross-country skiers during training have been reported to expend up to 9,000 kcal per day. One ultra-endurance runner expended an average of 10,750 kcal per day over a 5.2-day 600 mi (966 km) race!²⁹ Also, some competitive sports require adherence to rigid weight standards. Athletes who participate in these sports must closely monitor their weight and thus their caloric intake. Too often, this leads to poor nutrition, reliance on supplements and drugs, dehydration, and serious health risks. In addition, the dietary tactics used by some athletes to achieve excessive weight loss are of increasing concern because of the potential association with eating disorders, such as anorexia nervosa and bulimia nervosa.

Assessing Body Composition

Body composition refers to the body's tissue composition. [Figure 15.1](#) illustrates three models of body composition. The first two divide the body into its various chemical or anatomical components; the last one simplifies body composition into two components, the fat mass and the fat-free mass, which is the model used in this book. **Fat mass** is often discussed in terms of **relative body fat**, which is the percentage of the total body mass that is composed of fat. **Fat-free mass** simply refers to all body tissue that is not fat, including bone, muscle, organs, and connective tissue.

Assessment of body composition provides additional information to both the coach and the athlete, beyond the basic measures of height and weight. As an example, if the center fielder of a Major League Baseball team is 188 cm (6 ft 2 in.) tall and weighs 91 kg (200 lb), is he at his ideal playing weight? Knowing that 4.5 kg (10 lb) of a total weight of 91 kg (200 lb) is fat weight and that the remaining 86.5 kg (190 lb) is fat-free weight provides considerably more insight than knowing height and weight alone. In this example, only 5% of his body weight is fat, which is about as low as any athlete should go.

Armed with this knowledge, both athlete and coach realize that this athlete's body composition is ideal. They should not be concerned with weight loss, even though standard height–weight charts indicate that the athlete is overweight. However, another baseball player of the same height and weight who has 23 kg (50 lb) of fat would be 25% fat. This would constitute a problem for an elite athlete: He would be overfat.

In most sports, the higher the percentage of body fat, the poorer the performance. An accurate assessment of the athlete's body composition provides valuable insight into the weight that allows optimal performance. But how do we determine an athlete's body composition?

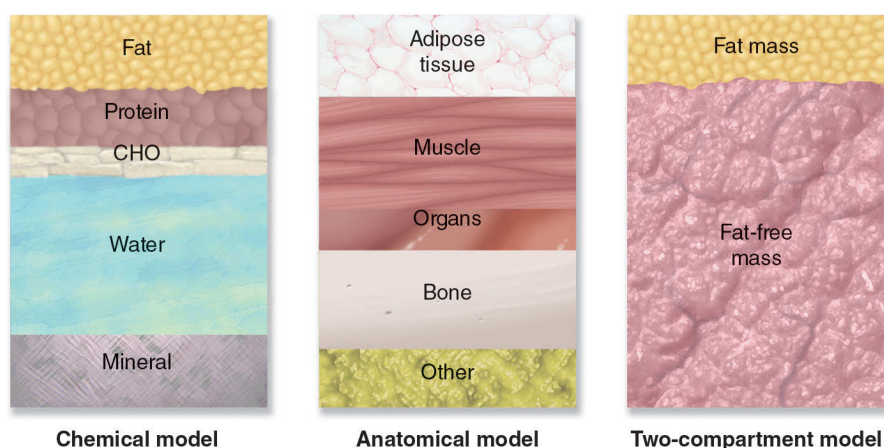


FIGURE 15.1 Three models of body composition (see text for description).

Adapted by permission from J.H. Wilmore, *Body Weight and Body Composition*, in *Eating, Body Weight, and Performance in Athletes: Disorders of Modern Society*, edited by K.D. Brownell, J. Rodin, and J.H. Wilmore (Philadelphia, PA: Lippincott, Williams, and Wilkins, 1992), 77-93.

Densitometry

Densitometry involves measuring the density of the athlete's body in order to assess body composition. Density (D) is defined as mass (M) divided by volume (V):

$$D_{\text{body}} = M_{\text{body}} / V_{\text{body}}$$

The mass of the body is the athlete's scale weight. Body volume can be obtained by several different techniques, but the most common is **hydrostatic weighing** (also called *underwater weighing*), in which the athlete is weighed while totally immersed in water. The difference

between the athlete's scale weight and underwater weight, when corrected for the density of water, equals the body's volume. This volume must be further corrected to account for air trapped in the body. The amount of air trapped in the gastrointestinal tract is small, difficult to measure, and usually ignored. The gas trapped in the lungs, however, must be measured because its volume is generally large.

Figure 15.2 shows the hydrostatic weighing technique. The density of the fat-free mass is higher than the density of water, while that of fat is lower than the density of water.

Densitometry has long been the technique of choice for assessing body composition. New techniques typically are often compared against densitometry to determine their accuracy. If body weight, underwater weight, and lung volume during underwater weighing are measured correctly, the resulting **body density** value is accurate. However, densitometry has its limitations. Densitometry's major weakness is in the conversion of body density to an estimate of relative body fat. Accurate estimates of the individual densities of fat mass and fat-free mass are required when the two-component model of body composition is used. The equation most often used to convert body density to an estimate of relative or percentage body fat is the standard equation of Siri:

$$\% \text{ Body fat} = (495 / D_{\text{body}}) - 450.$$

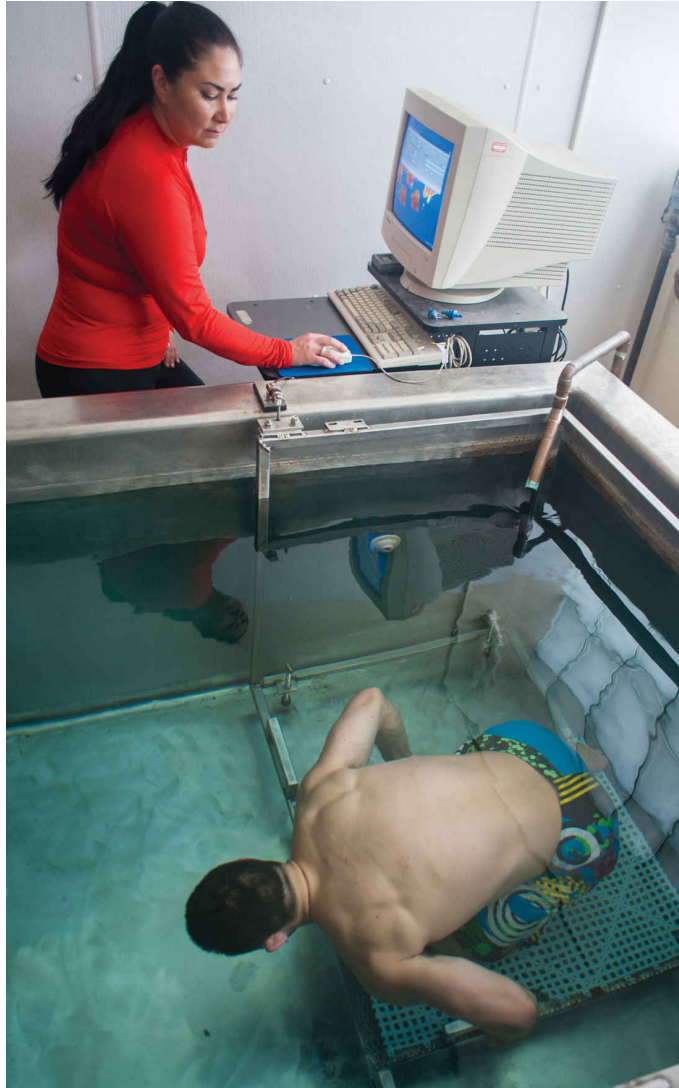


FIGURE 15.2 A simple hydrostatic weighing system (underwater weighing technique) to determine the density of the body.

This equation assumes that the densities of the fat mass and the fat-free mass are relatively constant in all people. Indeed, the density of fat at different sites is very consistent in the same individual and relatively consistent between people. The value generally used is 0.9007 g/cm. But determining the density of the fat-free mass (D_{FFM}), which the equation of Siri assumes is 1.100, is more problematic. To determine this density, we make assumptions that (1) the density of each tissue constituting the fat-free mass is known and remains constant and (2) each tissue type represents a constant proportion of the fat-free mass (e.g., we assume that bone always represents 17%

of the fat-free mass). Exceptions to either of these assumptions cause error when we convert body density to relative body fat.

Other Laboratory Techniques

Many other laboratory techniques are available for assessing body composition. These include radiography, computed tomography (CT), magnetic resonance imaging (MRI), hydrometry (for measuring total body water), total body electrical conductivity, and neutron activation. Most of these techniques are complex and require expensive equipment. None of them is likely to be used for the general assessment of athletic populations, so we will not discuss them further in this chapter; however, we do discuss CT in [chapter 22](#). Two other techniques hold considerable promise—dual-energy X-ray absorptiometry and air displacement.

Dual-energy X-ray absorptiometry (DEXA) evolved from the earlier single- and dual-photon absorptiometry techniques used between 1963 and 1984. The earlier techniques were used to estimate regional bone mineral content and bone mineral density, primarily in the spine, pelvis, and femur. The newer DEXA technique (see [figure 15.3](#)) allows the quantification of not only bone but also soft tissue composition. Furthermore, it is not limited to regional estimates but can provide total body estimates. Research to date suggests that DEXA provides precise and reliable estimates of body composition. DEXA is becoming more commonly accepted as the gold standard technique. The advantages of DEXA over the underwater weighing technique include the ability to estimate bone density and bone mineral content in addition to fat and fat-free mass. Furthermore, it is a passive technique whereby the athlete simply lies on a table during the scan, as opposed to having to be submerged underwater multiple times. The disadvantage is the cost of the equipment and technical support.

Air plethysmography, similar to hydrostatic weighing, is a densitometric technique. Volume is determined by air displacement rather than by water immersion. Commercial models of air plethysmographs have made this technique popular and widely available (see [figure 15.4](#)). The principle of operation is rather simple. It involves a closed chamber of room air at atmospheric pressure, which has a known volume. The individual to be tested opens the

chamber door, is seated in the chamber, then closes the chamber door, forming an airtight seal. The new volume of the air in the chamber is determined, which is then subtracted from the total volume of the chamber to provide an estimate of the person's volume.



FIGURE 15.3 (a) The dual-energy X-ray absorptiometry (DEXA) machine used to estimate bone density and bone mineral content as well as total body composition (fat mass and fat-free mass). (b) A regional scan of the body.



FIGURE 15.4 The Bod Pod air plethysmography device uses the air displacement technique to estimate total body volume.

Although this is a relatively simple technique for the subject, it requires considerable accuracy in controlling for changes in temperature, gas composition, and the subject's breathing while in the chamber. Studies have confirmed the accuracy of this technique under most conditions. It appears to provide a relatively precise measurement of body volume. Just as with the underwater weighing technique, one must still use the subject's body density in an equation to estimate relative body fat, recognizing the uncertainties of the D_{FFM} for that subject.

Field Techniques

Several field techniques are also available for assessing body composition. These techniques are more accessible than laboratory techniques because the equipment is less costly and cumbersome; they are techniques that therefore can be used more easily outside the laboratory by the coach, the trainer, or even the athlete.

Skinfold Fat Thickness

The most widely applied field technique involves measuring the **skinfold fat thickness** (see [figure 15.5](#)) at one or more sites and using the values obtained to estimate body composition. It generally is recommended that the sum of the measurements from three or more skinfold sites be used in a quadratic, curvilinear equation to estimate body density. A curvilinear equation more accurately describes the relationship between the sum of skinfold measurements and body density. Linear equations underestimate the density of lean people and overestimate the density of obese people. Skinfold fat thickness measurements that use quadratic equations provide reasonably accurate estimates of total body fat or relative fat.



FIGURE 15.5 Measuring skinfold fat thickness at the triceps skinfold site.

Bioelectric Impedance

Bioelectric impedance is a simple procedure introduced in the 1980s that takes just a few minutes to perform. Four electrodes are attached to the body—at the ankle, the foot, the wrist, and the back of the hand. Newer technology utilizes foot plate and handheld electrodes—as shown in [figure 15.6](#). An undetectable current is passed through the electrodes on one side of the body (hand and foot), while the other side receives the current flow. Electrical conduction through the tissues between the electrodes depends on the water and electrolyte distribution in that tissue. Fat-free mass contains almost all the body water and the conducting electrolytes, so

conductivity is much greater in the fat-free mass than in the fat mass. The fat mass has a much greater impedance, meaning that it is much more difficult for the current to flow through the fat mass. Thus, the amount of current flow through the tissues reflects the relative amount of fat contained in those tissues.

With the bioelectric impedance technique, measurements of the impedance, the conductivity, or both are transformed into estimates of relative body fat. Estimates of relative body fat based on bioelectric impedance highly correlate with body fat measurements obtained through hydrostatic weighing. However, the relative body fat in lean athletic populations tends to be overestimated with bioelectric impedance because of the nature of the equations used. Furthermore, hydration alters the bioimpedance and therefore must be tightly controlled.



FIGURE 15.6 A commercially available bioelectric impedance device marketed for assessing relative body fat.

In Review

- Knowing a person's body composition is more valuable for predicting performance potential than merely knowing height and weight.
- Densitometry is one of the best methods for assessing body composition and has long been considered the most accurate, although it does carry certain risks of error. It involves calculating the density of the athlete's body by dividing body mass by body volume, which is typically determined via hydrostatic weighing or air displacement. Body composition can be calculated, although there is some margin of error.

- Dual-energy X-ray absorptiometry, originally developed for estimating bone density and bone mineral content, is now capable of providing accurate estimates of not only total body composition—fat mass and fat-free mass—but also segmental body composition and bone mass.
 - Field techniques for assessing body composition include measuring skinfold fat thickness and bioelectric impedance. These techniques are less costly and more accessible to the athlete and the coach than are laboratory techniques.
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Body Composition, Weight, and Sport Performance

In certain sports (e.g., football and basketball), a large size traditionally has been associated with performance quality: The bigger the athlete, the better the performance. But big does not always mean better. In certain other sports, smaller and lighter are considered better for performance (e.g., gymnastics, figure skating, and diving). Yet this can be taken to extremes, compromising the athlete's health and performance. In the following sections, we consider how performance can be affected by weight and body composition.

Fat-Free Mass and Relative Body Fat

Rather than be concerned with total body size or weight, most athletes should be concerned specifically with fat-free mass. Maximizing fat-free mass is desirable for athletes involved in activities that require strength, power, and muscular endurance. But increased fat-free mass may be undesirable for the endurance athlete, such as a distance runner, for whom a higher fat-free mass is an additional load that must be carried and might impair the athlete's performance. This might also be true for the high jumper, long jumper, triple jumper, and pole-vaulter, who must maximize their vertical or horizontal distances or both. Additional weight, even though it is active fat-free mass, could decrease rather than facilitate performance in these events.

Relative body fat is a major concern of athletes. Adding more fat to the body just to increase the athlete's weight and overall size is generally detrimental to performance. This is true of all activities in which the body weight must be moved through space, such as

running and jumping. It is less important for more stationary activities, such as archery and shooting. Endurance athletes try to minimize their fat stores because excess weight has been proven to impair their performance.

Heavyweight weightlifters might be exceptions to the general rule that less fat is better. These athletes add large amounts of fat weight just before competition under the premise that the additional weight will lower their center of gravity and give them a greater mechanical advantage in lifting. The sumo wrestler is another notable exception to the theory that overall size is not the major determinant of athletic success. In this sport, the larger individual has a decided advantage; but even so, the wrestler with the higher fat-free mass should have the best overall success. Performance in swimming also seems to be an exception to this general rule. Body fat might provide some advantage to the swimmer because it improves buoyancy, which can reduce body drag in the water and reduce the metabolic cost of staying on the surface.

RESEARCH PERSPECTIVE 15.1

Exercise Type and Body Composition

Despite the well-established benefits of exercise on health and chronic disease, the precise role of exercise in weight loss is still controversial, likely resulting from the differences in outcome measures in the variety of studies that test whether regular exercise is associated with decreased body weight. Also, changes in outcome measures such as body weight or body mass index (BMI) may not reflect changes in body composition, which may be more likely to occur in response to adherence to prolonged exercise training.

Exercise interventions for weight loss or weight maintenance have often focused on aerobic exercise because of the higher energy expenditure as compared to resistance exercise. However, resistance exercise is associated with increased generalized functional capacity, which may affect total daily energy expenditure by increasing total physical activity. Furthermore, many exercise intervention trials have found that participants in an exercise program reduce their daily nonexercise activity or increase energy intake, both of which may minimize the potential weight reduction. Examining the effects of habitual exercise, rather than a prescribed intervention, may eliminate these compensatory metabolic and behavioral changes and enrich our scientific understanding of the long-term effects of exercise on body weight and composition.

In 2015, a team of researchers examined the effects of various self-selected exercise programs on measures of body composition.¹⁵ Researchers followed 430 young adults who were participants in the *Energy Balance Study*, a long-term observational study that examines determinants of weight change over the span of 1 year. All of the participants were healthy and had BMIs between 20 and 35 kg/m². At the start of the study, height, weight, fat mass, fat-free mass, and lean tissue mass were measured on all of the participants. Percent body fat was calculated and participants were classified into three groups: normal fat (%BF = 22±7), overfat (%BF = 30±8), and obese (%BF = 35±7). Every 3 months, the participants reported their habitual self-selected engagement (frequency and time) in different exercise types. Based on these reports, time spent engaging in endurance exercise, resistance exercise, and other exercise (e.g., sports, group fitness, or aerobics) was calculated for each participant. At the end of the study, 348 of the 430 original participants had provided sufficient data to be included in the 1-year analysis.

The authors found that neither total exercise nor specific exercise type had an impact on BMI. Resistance exercise increased lean body mass and decreased fat mass, while aerobic exercise decreased fat mass but had no effect on lean mass. All exercise types positively affected lean body mass in the normal-fat participants. Interestingly, in overfat and obese participants, fat mass was reduced by increasing resistance exercise but was not changed with increasing aerobic exercise. The research team concluded that despite the limited effects on BMI, exercise was associated with beneficial changes in body composition such as increased lean mass in normal-fat participants and reduced fat mass in overfat and obese participants. Additionally, adults with excess body fat may benefit more from resistance exercise training, contrary to the commonly held belief that they should focus on aerobic exercise training.

Weight Standards

Weight standards are now used in several sports to ensure that athletes have the optimal body size and composition for maximal performance. Unfortunately, this is not always the result.

Theoretically, the elite athlete's genetic foundation and years of intense training have combined to provide the ultimate athletic profile for the given sport. These elite athletes set the standards toward which others aspire. However, this can be misleading, as we see in [figure 15.7](#).³⁶ The figure displays the percentage body fat values of elite female track-and-field athletes. If we look just at the distance runners, many of the best were below 12% body fat and the top two had only about 6% fat. However, one of the best distance runners in

the United States at that time had 17% body fat. Furthermore, one of the women in this study had a relative body fat of 37%, and she set the best time in the world for the 50 mi (80 km) run within 6 months of her evaluation! Neither of these women would have been successful if she had been forced to decrease her weight to achieve 12% body fat or lower.

Furthermore, weight standards have been abused over the years by coaches, athletes, and parents who have adopted the philosophy that if small weight losses improve performance a little, then major weight losses should improve it even more.

Risks Associated with Severe Weight Loss

Many schools, districts, or state- and national-level organizations organize sports (e.g., wrestling) on the basis of size, with weight as the predominant factor. Athletes in these sports often attempt to achieve the lowest possible weight to gain an advantage over opponents. In so doing, many have jeopardized their health. In the following sections, we examine a few of the consequences of severe weight loss in athletes, both male and female.

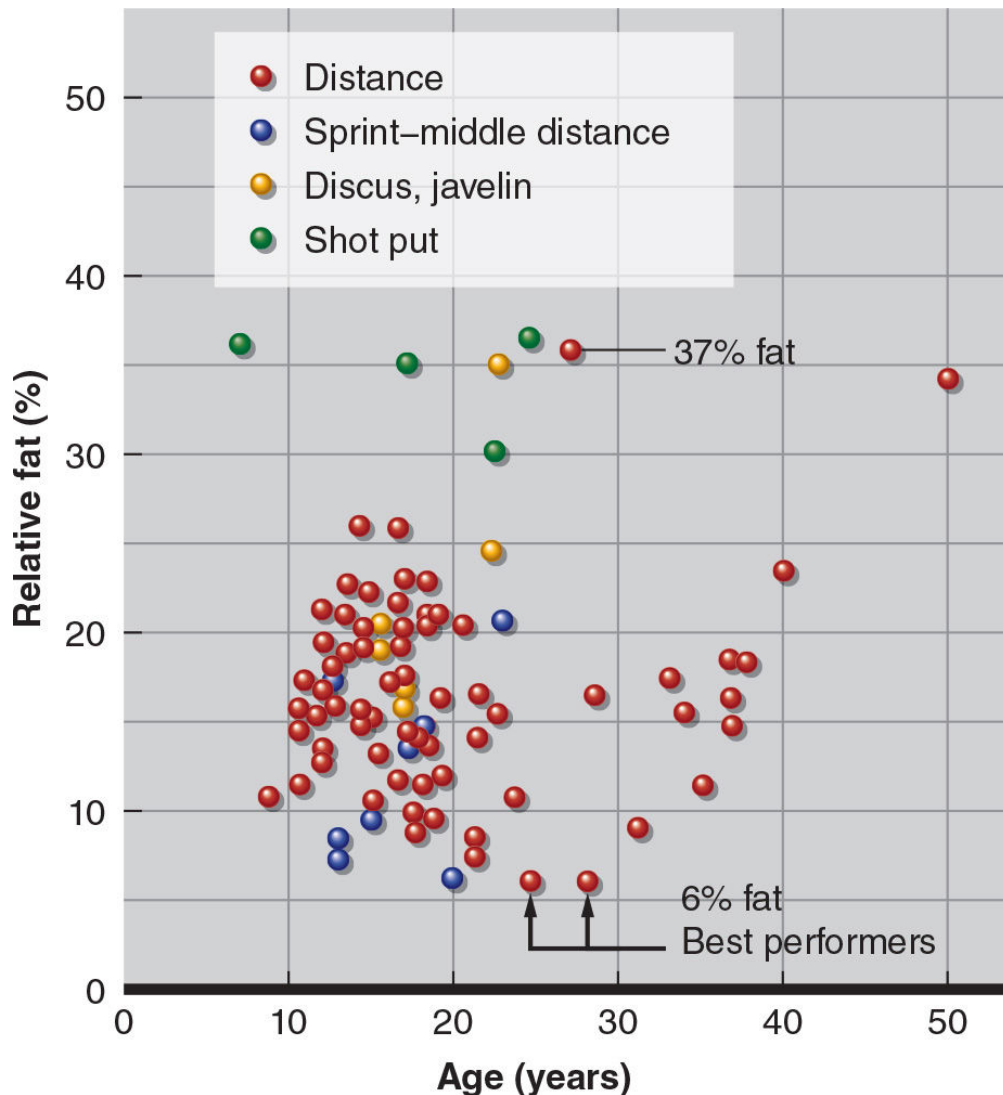


FIGURE 15.7 Relative body fat in elite female track-and-field athletes (see text for explanation).
Data from Wilmore, Brown, and Davis (1977).

Dehydration Fasting or very low-calorie diets lead to large amounts of weight loss, primarily through dehydration. As we discuss later in this chapter, for every gram of carbohydrate stored, there is an obligatory gain of 2.6 g of water. When carbohydrate is used for energy, that water is lost. With fasting and very low-calorie diets, carbohydrate stores are substantially depleted during the first few days. This results in a significant loss of weight attributable to the loss of body water.

Additionally, athletes trying to make weight might exercise in rubberized sweat suits, sit in steam and sauna baths, chew on towels to lose saliva, and minimize their fluid intake. Such severe water

losses compromise kidney and cardiovascular function and are potentially dangerous. Losses of as little as 2% of the athlete's weight through dehydration can impair performance, even skilled performance in intermittent, high-intensity sports like tennis, soccer, and basketball.¹⁷

Fatigue Pushing body weight too low can have major repercussions. When weight drops below a certain level, the athlete is likely to experience performance decrements and increased incidence of illness and injury. The performance decrements can be attributable to many factors, including the prolonged feelings of fatigue that often accompany excessive weight loss. The causes of this fatigue have not been established, but there are several likely possibilities. This type of fatigue could be attributed to substrate depletion. Both neural and hormonal components seem to be involved. In some cases, the balance between the sympathetic nervous system and the parasympathetic system may be altered in favor of the latter. In addition, effects of severe weight loss on the hypothalamus can impair immune function.

The Female Athlete Triad The constant attention given to achieving and maintaining a prescribed weight goal, particularly if the weight goal is inappropriate, can lead to energy imbalances, especially in female athletes. The precipitating cause of these disturbances is the failure of the athlete to consume an adequate number of calories to meet the energy expenditure needs associated with exercise training.¹ As a result, the athlete is in a state of energy deficiency or low energy availability. A female athlete who experiences energy imbalance for prolonged periods (low energy availability compared to high energy expenditure associated with training, for example) is prone to disordered eating, menstrual dysfunction, and bone mineral loss.¹ Together, these disorders are referred to as the female athlete triad. These important problems are discussed in more detail in [chapter 19](#).

Establishing Appropriate Weight Standards

The potential for abuse of weight standards is clearly established. Thus, it is critically important to properly set weight standards. Body weight standards for sports should be based on an athlete's body composition and not purely the body mass. Thus, establishing weight

standards should translate into establishing standards of relative body fat for each sport and, where appropriate, for each event within a sport. With this in mind, what is the recommended relative body fat for an elite athlete in any given sport? For each sport, an optimal range of values for relative body fat should be established, outside of which the athlete's performance is likely impaired. And because fat distribution shows definite sex differences, the weight standards should be sex specific. Representative ranges for men and women in various sports are presented in [table 15.1](#). In most cases, these values represent the elite athletes in those sports.

TABLE 15.1 Ranges of Relative Body Fat Values for Male and Female Athletes in Various Sports

Group or sport	% fat	
	Men	Women
Baseball or softball	8-14	12-18
Basketball	6-12	10-16
Bodybuilding	5-8	6-12
Canoeing or kayaking	6-12	10-16
Cycling	5-11	8-15
Fencing	8-12	10-16
Football	5-25	
Golf	10-16	12-20
Gymnastics	5-12	8-16
Horse racing (jockey)	6-12	10-16
Ice or field hockey	8-16	12-18
Orienteering	5-12	8-16
Pentathlon	*	8-15
Racquetball	6-14	10-18
Rowing	6-14	8-16
Rugby	6-16	*
Skating	5-12	8-16
Skiing (alpine and Nordic)	7-15	10-18
Ski jumping	7-15	10-18
Soccer	6-14	10-18
Swimming	6-12	10-18
Synchronized swimming		10-18
Tennis	6-14	10-20
Track and field, field events	8-18	12-20
Track and field, running events	5-12	8-15
Triathlon	5-12	8-15
Volleyball	7-15	10-18
Weightlifting	5-12	10-18
Wrestling	5-16	

*Data not available.

The recommended values may not be appropriate for all athletes who engage in a specific activity. The existing techniques for measuring body composition include inherent errors, as we discussed earlier in this chapter. Even with the best laboratory techniques, measurement of body composition is associated with a 1% to 3% error or higher. In addition, we must understand the concept of

individual variability. Not every female distance runner will have her best performance at 12% body fat or lower. While some will improve performance with these low values, others won't be able to get down to such low relative fat values, or they will find that their performance starts to decline before they reach the suggested values. For these reasons, a range of values should be set for men and women in specific activities, recognizing individual variability, methodological error, and sex differences.

Achieving Optimal Weight

Many athletes find themselves considerably above their assigned playing weight with only a few weeks remaining before they report to training camp. Consider the NFL lineman introduced at the beginning of this chapter. He had to weigh 149.7 kg (330 lb) to achieve peak performance and receive a financial incentive but weighed 157.3 kg (346 lb). If that scenario took place 1 month before the competitive season started, he would have some work to do to reach his goal. Exercise alone is of little value because he would need 6 to 8 months or more to lose this much weight through that means. Will this athlete accomplish his goal?

VIDEO 15.1 Presents Louise Burke on strategies for in-season diet changes for athletes who need to lose or gain weight.



Avoiding Fasting and Crash Dieting

Consider a wrestler who decides that he wants to wrestle at a much lower weight class in order to be more competitive. He wants to lose 3 kg (6.6 lb) per week for the next 4 weeks, so he decides to embark

on a crash diet, selecting whatever diet is in vogue, knowing that a person can lose weight quickly with such diets. But much of this weight loss would be from body water and very little from stored fat. Several studies have shown that substantial weight losses occur with very low-calorie diets (of 500 kcal per day or less) over the first several weeks, but that of the weight lost, more than 60% comes from the body's fat-free tissue and less than 40% from fat depots.

While much of the player's weight is lost from water stores, a substantial amount of protein is lost as well. Also, most crash diets are based on a major reduction in carbohydrate intake. This reduced intake is insufficient to supply the body's needs for carbohydrate, and as a result the body's carbohydrate stores become depleted. Because water storage accompanies carbohydrate storage, the water stores are also reduced as the carbohydrate stores diminish, as discussed earlier in this chapter.

In addition, the body relies more heavily on free fatty acids for energy because its carbohydrate stores are depleted. As a result, ketone bodies, a by-product of fatty acid metabolism, accumulate in the blood, causing a condition known as ketosis. This condition further increases water loss, especially during the first week of the diet. Athletes who have attempted this ill-advised shortcut to rapid weight loss have lost substantial weight, but, because much of the weight loss was from the fat-free mass, their performance was severely compromised.

Optimal Weight Loss: Decreasing Fat Mass and Increasing Fat-Free Mass

The sensible approach to reducing body fat stores is to combine moderate dietary restriction with increased physical activity. Athletes should work toward achieving their goal weight slowly, losing no more than 1 kg (less than 2.2 lb) per week. Losing more weight than that per week leads to losses in fat-free mass. The rate of this loss should be reduced still more, or the weight loss program terminated, if performance is affected or if medical symptoms are noted.

Decreasing caloric intake by 200 to 500 kcal per day will allow weight losses of about 0.5 kg (1.1 lb) per week, particularly if combined with a sound exercise program. This is a realistic goal, and such losses add up to a substantial weight loss over time. When

trying to reduce weight, athletes should consume their total daily calories over at least three meals per day. Many athletes make the mistake of eating only one or two meals per day, skipping breakfast, lunch, or both, and then consuming a large dinner. Research in animals has shown that, given the same number of total calories, the animals that eat their daily food ration in one or two meals gain more weight than those that nibble their ration throughout the day. Human research is less clear.

The purpose of weight loss programs is to lose body fat, not fat-free mass. Therefore, the combination of diet and exercise is the preferred approach. Combining increased activity with caloric reduction prevents any significant loss of fat-free mass. Resistance training promotes gains in fat-free mass, and both resistance training and endurance training promote loss of fat mass. To lose weight, athletes should combine a moderate resistance and endurance training program with modest caloric restriction.

In Review

- Maximizing fat-free mass is desirable for athletes in sports that require strength, power, and muscular endurance but could be a hindrance to endurance athletes, who must be able to move their total body mass for extended periods, and jumpers, who must move their body mass vertically or horizontally for distance.
- The degree of fatness has more influence on performance than does total body weight. In general, the greater the relative body fat, the poorer the performance. Possible exceptions include heavyweight weightlifters, sumo wrestlers, and swimmers.
- Severe weight loss in athletes can cause potential health problems, such as dehydration, chronic fatigue, disordered eating, menstrual dysfunction, and bone mineral disorders.
- Body weight standards should be based on body composition and should emphasize relative body fat rather than total body mass.
- For each sport, a range of values should be established, recognizing the importance of individual variation, methodological error, and sex differences.
- When severe (very low calorie) diets are followed, much of the weight loss that occurs is from water, not fat.
- Most severe diets limit carbohydrate intake, depleting carbohydrate stores. Water is lost along with the carbohydrate, exacerbating the problem of dehydration. Also,

the increased reliance on free fatty acids can lead to ketosis, which further increases water loss.

- The combination of diet and exercise is the preferred approach to optimal weight loss.
 - Athletes should lose no more than about 1 kg (about 2 lb) per week until reaching the upper end of the desired weight range. After that, weight loss should be less than 0.5 kg (1 lb) per week until goal weight is reached. More rapid weight losses result in loss of fat-free mass. People can accomplish weight loss at the recommended rate by reducing dietary intake 200 to 500 kcal per day, especially in combination with a sound exercise program.
 - For fat loss, moderate resistance and endurance training is most effective. Resistance training promotes gains in fat-free mass.
-

Classification of Nutrients

We now turn our attention to nutrition aspects of preparing the athlete for optimal performance. As we will see, it is important to maintain a diet that will provide general health benefits, maintain an appropriate weight and body composition, and maximize athletic performance.

A person's diet should contain a relative balance of carbohydrate, fat, and protein. Of the total calories consumed, the recommended balance for most people is

- carbohydrate—55% to 60%;
- fat—no more than 35% (less than 10% saturated); and
- protein—10% to 15%.

Interestingly, this recommended percentage distribution of total calories consumed appears to be optimal for both athletic performance and health. However, when planning diets for athletes, using grams per kilogram of body weight for carbohydrate and protein results in a more precise prescription for training and competition needs.²⁷ A similar distribution of caloric intake is recommended for the prevention of cardiovascular disease, diabetes, obesity, and cancer, as discussed in greater detail later in this chapter. Although all foods ultimately can be broken down to carbohydrate, fat, or protein, these nutrients are not all that the body needs. Food can be

categorized into six classes of nutrients, each with specific functions in the body:

- Carbohydrate
- Fat
- Protein
- Vitamins
- Minerals
- Water

The following discussion examines the physiological importance to the athlete of each class of nutrients.

RESEARCH PERSPECTIVE 15.2

Meal Timing and the Aerobic Exercise Window

In 1982, a study published in the *American Journal of Physiology* demonstrated that moderate-intensity physical activity (treadmill walking) that began 30 min after a meal rapidly returned blood glucose levels to fasting levels.²⁴ Over the decades that have passed since that finding, research has shown that blood glucose concentrations are sensitive to multiple exercise conditions, including timing, intensity, duration, frequency, and sequencing of exercise modes. A large number of studies have examined exercise and meal timings; however, a wide variety in exercise intensities, mode, and meal timings have led to conflicting reports. For example, light to moderate exercise before a meal raises postprandial glucose, but similar activity after a meal lowers blood glucose. However, changing to a high-intensity interval exercise bout gives the opposite result, with premeal exercise improving the glycemic response. Despite these conflicting results, the consensus is that postmeal exercise of moderate intensity is superior to premeal exercise if your goal is to limit hyperglycemia. Moderate exercise during the mid-postprandial period consistently lowers the postmeal spike in blood glucose, and exercise starting 30 min after the first bite into the meal shows the most effective blunting of the glucose surge. These data are especially important in meal timing and exercise plans for individuals with diabetes, who are challenged to manage glycemia closely.¹⁰

Building on these data surrounding meal consumption and exercise timing, a 2015 study examined the role of postexercise protein consumption on skeletal muscle protein synthesis rate and recovery after aerobic exercise in 12 aerobically trained men.²⁸ On three separate trials, each subject completed high-intensity cycling then ingested either (1) a protein–carbohydrate blend with 5 g of leucine, (2) a protein–carbohydrate blend with 15 g of leucine, or

(3) a non-nitrogenous, isocaloric control beverage. The leucine doses were chosen based on previous studies showing that 15 g of leucine improved recovery and increased muscle protein synthesis after resistance exercise, as well as to see if aerobic athletes could confer similar benefits with lower doses. The research team collected muscle biopsies to determine muscle protein fractional synthetic rate during recovery. They also collected blood samples to measure plasma amino acids throughout recovery.

The researchers found that the beverages containing protein increased muscle protein fractional synthetic rate compared to the control. The addition of 5 mg leucine increased muscle protein fractional synthetic rate by 33% and represented near-maximum stimulation for protein synthesis. Further increasing leucine did not increase muscle protein synthesis above what was observed with the lower dose beverage. Plasma amino acids and leucine were decreased during recovery following the control beverage, but increased in a dose-dependent manner with ingestion of the protein–leucine beverages. The study concluded that ingestion of a protein-containing meal improves muscle synthesis and recovery following a bout of aerobic exercise; however, the addition of 5 mg of leucine is sufficient to reach maximal fraction protein synthesis. Although additional supplementation after that does increase plasma amino acids, it does not result in additional muscle synthesis. To address this, future studies should attempt to determine if these customized protein–leucine recovery beverages also translate to improvements in performance.

Carbohydrate

A carbohydrate (CHO) is classified as either a monosaccharide, a disaccharide, or a polysaccharide. Monosaccharides are the simple one-unit sugars, such as glucose, fructose, and galactose, that cannot be reduced to a simpler compound. Disaccharides (such as sucrose, maltose, and lactose) are made up of two monosaccharides. For example, sucrose (table sugar) consists of glucose and fructose. Oligosaccharides are short chains of 3 to 10 monosaccharides linked together. Polysaccharides are composed of large chains of linked monosaccharides. Glycogen is the energy-providing polysaccharide found in animals, including man, and is stored in muscle and liver. Starch and fiber are the two plant polysaccharides and are commonly referred to as complex carbohydrates. Simple carbohydrates are carbohydrates derived from processed foods or foods high in sugar. All carbohydrates must be broken down to monosaccharides before the body can use them.

Carbohydrate serves many functions in the body:

- It is a major energy source, particularly during high-intensity exercise.
- Its presence regulates fat and protein metabolism.
- The nervous system relies exclusively on carbohydrate for energy.
- Muscle and liver glycogen are synthesized from ingested carbohydrates.

Major sources of carbohydrate in the diet include grains, fruits, vegetables, milk, and concentrated sweets. Refined sugar, syrup, and cornstarch are nearly pure carbohydrates, as are candy, honey, jellies, and molasses. Some soft drinks contain few if any nutrients other than carbohydrates.

Carbohydrate Consumption and Glycogen Storage

The body stores excess carbohydrate, primarily in the muscles and liver, as glycogen. Because of this, carbohydrate consumption directly influences muscle glycogen storage and the ability to train and compete in endurance events. As shown in [figure 15.8](#), athletes who trained intensely over 3 consecutive days and ate a low-carbohydrate diet (40% of total calories) experienced a progressive decrease in muscle glycogen.¹² When these same athletes consumed a high-carbohydrate diet (70% of total calories), their muscle glycogen levels recovered almost completely within the 22 h between training bouts.

Early studies demonstrated that when men ate a diet containing a normal amount of carbohydrate (about 55% of total calories ingested), their muscles stored about 100 mmol of glycogen per kg of muscle. One study showed that diets containing less than 15% carbohydrate led to storage of only 53 mmol/kg, but carbohydrate-rich diets (60%-70% CHO) led to storage of 205 mmol/kg. When subjects exercised to exhaustion at 75% of their maximal oxygen uptake, their exercise times were proportional to the amount of muscle glycogen stored before the test, as shown in [figure 15.9](#).

Most studies have shown that glycogen storage replacement is not determined simply by carbohydrate intake. Exercise with an eccentric (muscle lengthening) component, such as running and weightlifting, can induce some muscle damage and impair glycogen resynthesis. In these situations, muscle glycogen levels can appear quite normal

during the first 6 to 12 h after exercise, but glycogen resynthesis slows or stops completely as muscle repair begins.

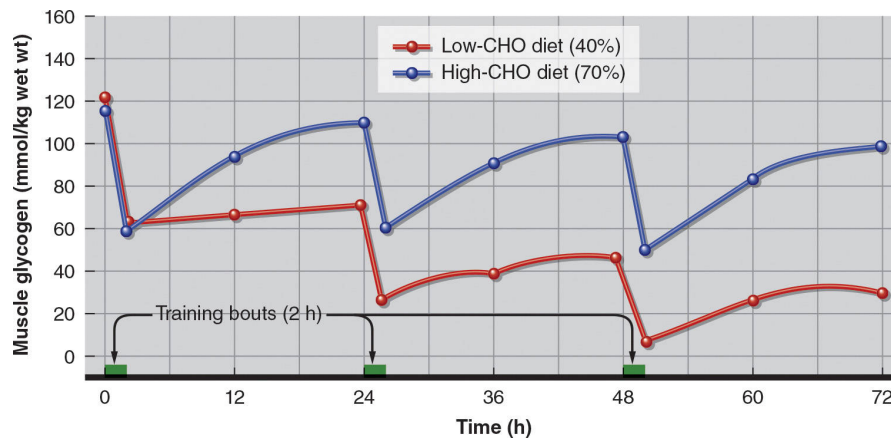


FIGURE 15.8 The influence of dietary carbohydrate (CHO) on muscle glycogen stores during repeated days of training. Note that when a low-CHO diet was consumed, muscle glycogen gradually declined over the 3 days of study, whereas the CHO-rich diet was able to return the glycogen to near normal each day.

Data from Costill and Miller (1980); Costill et al. (1971).

The precise cause for this response is unknown, but conditions within the muscle could inhibit muscle glucose uptake and glycogen storage. For example, within 12 to 24 h after intense eccentric exercise, damaged muscle fibers are infiltrated with inflammatory cells (leukocytes, macrophages) that remove cellular debris resulting from damage to the cells' membranes (see [chapters 5](#) and [14](#)). This repair process can require a significant amount of the blood glucose, reducing the amount of glucose available for resynthesizing muscle glycogen. In addition, some evidence suggests that eccentrically exercised muscle is less sensitive to insulin, which would limit muscle fiber uptake of glucose. Perhaps future studies will more fully explain why eccentric-type activities delay glycogen storage. But for now, we can only observe that glycogen recovery from various forms of exercise can differ and that this should be considered for optimal diet, training, and competition.

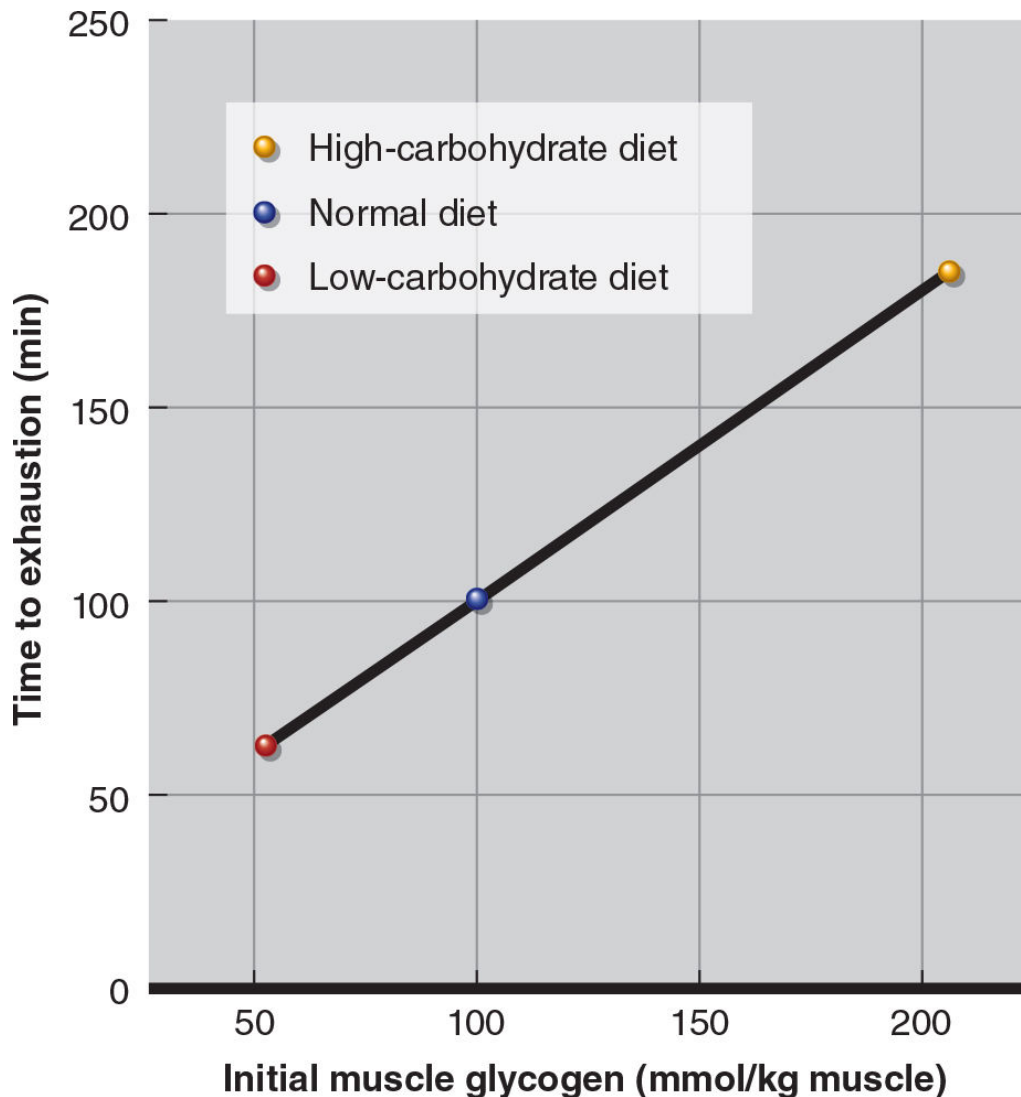


FIGURE 15.9 The relation between preexercise muscle glycogen content and exercise time to exhaustion. The exercise time to exhaustion and muscle glycogen were nearly four times greater when the subjects ate a carbohydrate-rich diet than when the diet was composed mostly of fat and protein.

Carbohydrate is the primary fuel source for most athletes and should constitute at least 50% of their total caloric intake. For endurance athletes, carbohydrate intake as a percentage of total caloric intake might need to be higher: 55% to 65%. However, most important is the total number of grams of carbohydrate ingested. It appears that athletes need from 3 to 12 g/kg of body weight per day in order to maintain glycogen stores. This wide range is necessary to account for the training intensity and total daily energy expenditure, the person's sex, and environmental conditions. For example, during periods of moderate-intensity training, 5 to 7 g/kg per day should be

adequate. However, with long-duration high- and extremely high-intensity training, intake should be increased to 6 to 10 g/kg per day and 8 to 12 g/kg per day, respectively.^{9,23}

The Glycemic Index

It has long been thought that the rapid increase in blood sugar concentration (hyperglycemia) with the intake of carbohydrate usually is associated with simple carbohydrates, such as glucose, sucrose, fructose, and high-fructose corn syrup. However, this is not always the case. Scientists have discovered that the glycemic response (i.e., increase in blood sugar) to carbohydrate intake varies considerably for both simple and complex carbohydrates. This has led to the use of what has been termed the *glycemic index* of foods (GI). The ingestion of glucose or white bread leads to a rapid increase in blood sugar. This response is used as a standard and has been arbitrarily assigned a GI of 100. The glycemic response for all other foods is referenced against the response for glucose or white bread, using 50 g of both the test food and glucose or white bread as the standard. The GI is calculated as follows: $GI = 100 \times (\text{blood glucose response over 2 h to 50 g of test food} / \text{blood glucose response over 2 h to 50 g of glucose or white bread})$. Three categories of GI have been established:²³

- High glycemic index foods (GI >70) such as sport drinks, jelly beans, baked potato, French fries, popcorn, cornflakes, Corn Chex, and pretzels
- Moderate glycemic index foods (GI 56-70) such as pastry, pita bread, boiled white rice, bananas, most popular soft drinks, and regular ice cream
- Low glycemic index foods (GI ≤55) such as white boiled spaghetti, kidney and baked beans, milk, grapefruit, apples, pears, peanuts, M&M's, and yogurt

Food items were classified according to the 2002 International Table of Glycemic Index and Glycemic Load Values.

While the GI is a useful tool for rating foods, it is not without controversy. First, the GI for a given food can vary considerably between individuals as well as between similar foods. Second, some complex carbohydrates have high GIs. Third, adding small amounts

of fat to a high-GI carbohydrate can greatly reduce the GI of that food. Finally, GI values differ substantially depending on whether glucose or white bread is used as the reference food, with white bread producing substantially higher values.²³ An additional index has been proposed that might be of importance during exercise. The glycemic load (GL) considers both the GI and the amount of carbohydrate (CHO) in a single serving and is calculated as follows: $GL = (GI \times CHO, g)/100$.

Before exercise, low-GI foods would be preferred to reduce the likelihood of hyperinsulinemia. However, high-GI foods should be an advantage during exercise by helping maintain blood glucose levels. This should also be the case during recovery from intense and prolonged exercise, since the higher blood sugar level should increase muscle and liver glycogen storage.

Carbohydrate Intake and Performance

As noted earlier, muscle glycogen provides a major source of energy during exercise. Muscle glycogen depletion has been shown to be a major cause of fatigue and ultimate exhaustion in high-intensity exercise of short duration or in moderate-intensity exercise lasting more than an hour. This is clearly illustrated in [figure 15.10](#), which shows the marked depletion of muscle glycogen at very high intensities (150% and 120% of $\dot{V}O_{2max}$) for durations of less than 30 min, as well as at lower intensities (83%, 64%, and 31% of $\dot{V}O_{2max}$) for durations of 1 to 3 h.

On the basis of these results, scientists speculated that loading the muscle with extra glycogen before starting exercise should enhance performance. Studies in the 1960s demonstrated that men who ate a carbohydrate-rich diet for 3 days stored nearly twice their normal amounts of muscle glycogen.⁴ When they were asked to exercise to exhaustion at 75% of $\dot{V}O_{2max}$, their exercise times significantly increased (see [figure 15.9](#)). This practice, called **glycogen loading** or **carbohydrate loading**, is widely used by distance runners, cyclists, and other athletes who must perform exercise for several hours. We discuss this practice in greater detail later in this chapter.

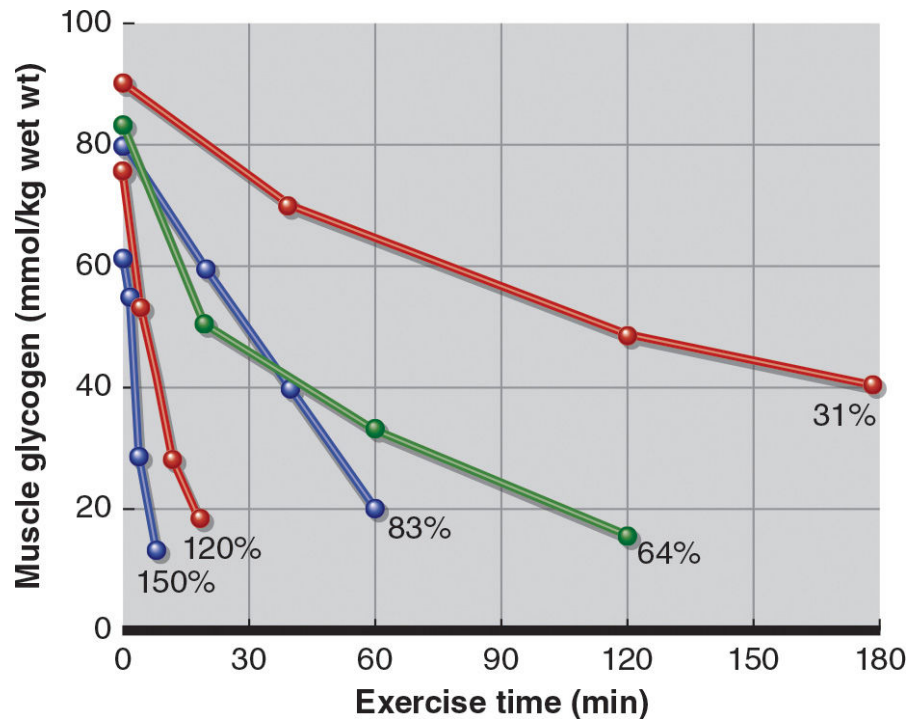


FIGURE 15.10 The influence of exercise intensity (31%, 64%, 83%, 120%, and 150% of $\dot{V}O_{2max}$) on the reduction in muscle glycogen stores. At relatively high intensities, the rate of muscle glycogen use is extremely high compared to that at the moderate and lower intensities.

Adapted by permission from A. Jeukendrup and M. Gleeson, *Sport Nutrition: An Introduction to Energy Production and Performance* (Champaign, IL: Human Kinetics, 2004). Original data from Gollnick, Piehl, and Saltin (1974).

Blood glucose concentrations become low (hypoglycemia) during exhaustive high-intensity, long-duration exercise, and this might contribute to fatigue. Numerous studies have shown that subjects' performances improve when they are given carbohydrate feedings during exercise lasting 1 to 4 h. Comparisons of subjects receiving carbohydrate feedings and those receiving placebos generally reveal no performance differences during the early stages of exercise, but during the final stages, performance is greatly improved with carbohydrate feedings.

Although all of the precise mechanisms by which carbohydrate feedings improve performance are not fully understood, maintaining blood glucose near normal concentrations allows the muscles to obtain more energy from this source. Carbohydrate feedings during exercise generally do not spare muscle glycogen use. Instead, they may preserve liver glycogen or even promote glycogen synthesis during exercise, enabling the exercising muscles to rely more on

blood glucose for energy late in the exercise. Carbohydrate feedings might also enhance central nervous system function, reducing the perception of effort. Endurance performance (more than 1 h) can be enhanced when carbohydrate is consumed within 5 min before the exercise begins, more than 2 h before exercise (such as during the precompetition meal), and at frequent intervals during the activity.

VIDEO 15.2 Presents Asker Jeukendrup on carbohydrate intake during exercise.



An athlete should use caution when ingesting carbohydrate-rich foods during the period from 15 to 45 min before exercise, because this could cause hypoglycemia shortly after exercise begins, which could lead to early exhaustion by depriving the muscle of its primary energy sources. Carbohydrate ingested during that period stimulates insulin secretion, elevating insulin when the activity begins. In response to the elevated insulin level, glucose uptake by the muscles reaches an abnormally high rate, leading to hypoglycemia and early fatigue ([figure 15.11](#)). Not everyone experiences this reaction, but sufficient evidence indicates that high-GI carbohydrates (those that cause a large increase in blood insulin) should be avoided or moderated in the period from 15 to 45 min before exercise.

Why don't carbohydrate feedings during exercise produce the same hypoglycemic effects observed with preexercise feedings? Sugar intake during exercise results in smaller increases in both blood glucose and insulin, lessening the threat of an overreaction that leads to a sudden decrease in blood glucose. This finer control of blood glucose during exercise might be caused by increased muscle

fiber permeability that decreases the need for insulin, or insulin-binding sites may be altered during muscular activity. Regardless of the cause, carbohydrate intake during exercise appears to supplement the carbohydrate supply needed for muscular activity.

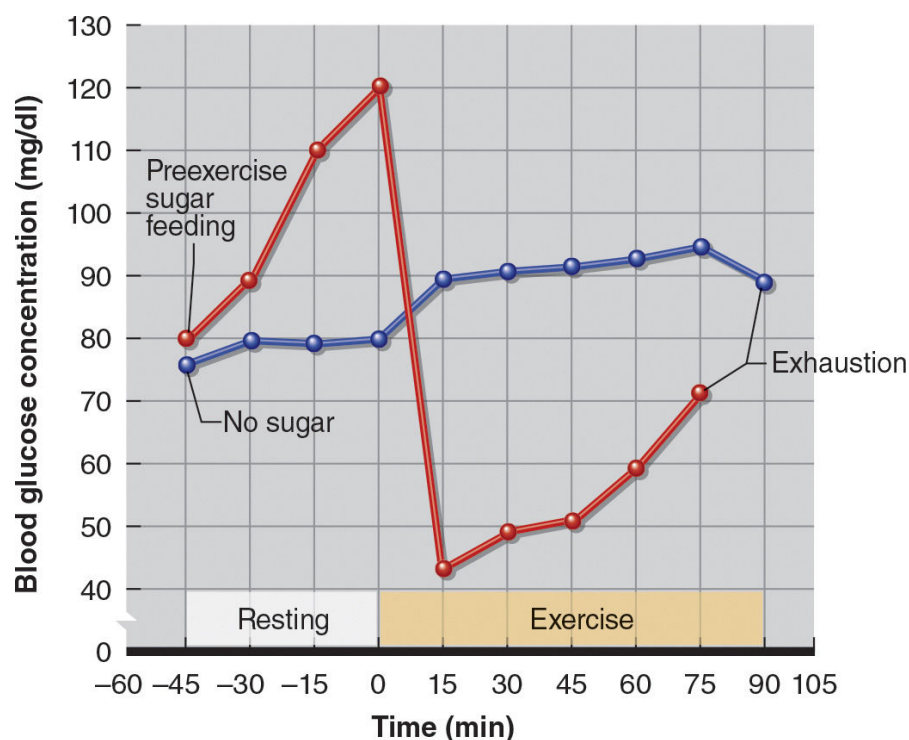


FIGURE 15.11 The effects of preexercise carbohydrate (sugar) feeding on blood glucose levels during exercise. Note the decrease in blood glucose to hypoglycemic levels with sugar feeding 45 min before exercise. Also, subjects during the sugar-feeding trial were unable to complete the full 90 min at 70% of $\dot{V}O_{2\max}$, achieving only 75 min.

Based on Costill et al. (1977).



Finally, it is important to consume carbohydrate immediately after high-intensity and long-duration exercise during which carbohydrate stores have been reduced or depleted. Rates of glycogen resynthesis are very high during the first 2 h of recovery and progressively decrease thereafter. In a study by Ivy and colleagues,²¹ cyclists exercised continuously for 70 min on a cycle ergometer on two

separate occasions, a week apart, at moderate to high work rates to deplete the active muscles' glycogen stores. During one trial, a 25% carbohydrate solution was ingested immediately after exercise, while in the other trial the solution was ingested after 2 h of recovery. In the trial in which the solution was provided immediately after exercise, glycogen storage rates during the first 2 h were three times higher compared to the trial in which the solution was not provided until after 2 h of recovery. The storage rates were the same for the two trials during the second 2 h (see [figure 15.12](#)). More recently, it has been shown that adding protein to the carbohydrate supplement enhances the replenishment of muscle glycogen stores during the recovery period. Adding protein to the carbohydrate supplement maximizes glycogen synthesis with less frequent supplementation and less carbohydrate.²³ Further, it also appears to stimulate muscle tissue repair.

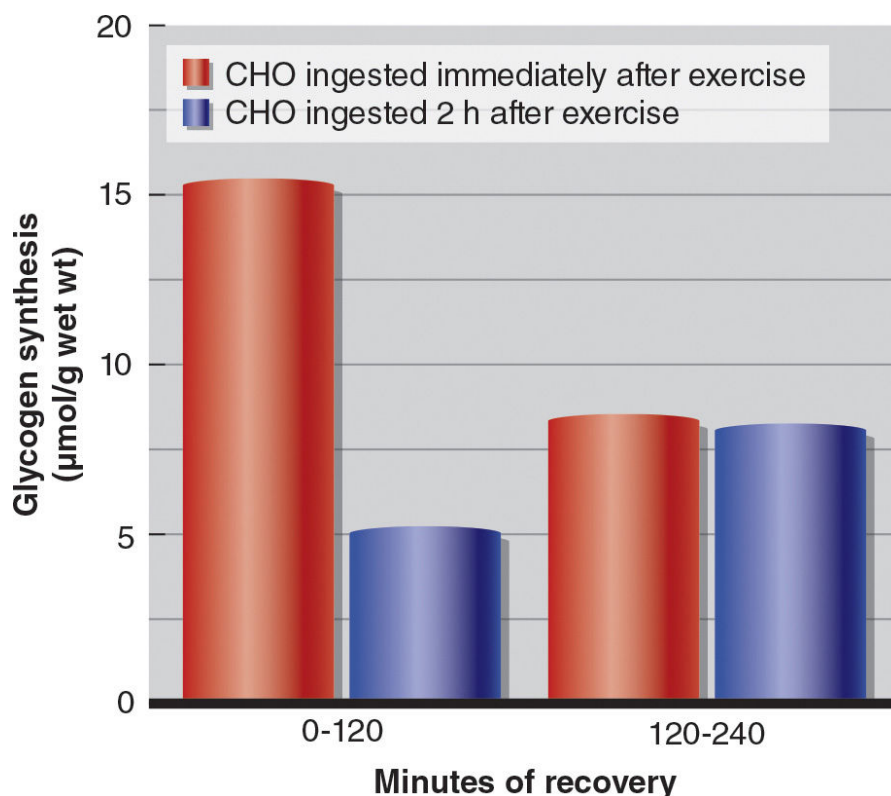


FIGURE 15.12 Replenishment of muscle glycogen stores after 70 min of muscle glycogen-depleting exercise using two different regimens of carbohydrate replacement. In the trial in which the carbohydrate solution was provided immediately following exercise (left), muscle glycogen storage was three times higher during the first 2 h of recovery compared to the other trial, in which the carbohydrate solution was not given until after 2 h of recovery (right). There was no difference in muscle glycogen storage during the next 2 h.

The importance of maximizing carbohydrate storage in the liver and muscles before exercise, and of providing carbohydrate during and immediately following exercise, has led food and nutrition companies to develop products to meet these needs, including sports drinks, as discussed at the end of this chapter.

Fat

Fat, also termed lipid, is a class of organic compounds with limited water solubility. It exists in the body in many forms, such as triglycerides, free fatty acids (FFAs), phospholipids, and sterols. The body stores most of its fat as triglycerides, composed of three molecules of fatty acids and one molecule of glycerol. Triglycerides are our most concentrated source of energy.

Excessive dietary fat, especially cholesterol and triglycerides, plays a major role in cardiovascular disease (discussed in [chapter 21](#)), and excessive fat intake also has been linked to other diseases such as cancer, diabetes, and obesity. But despite the negative effects of too much fat intake, fat is necessary in the diet and serves many vital functions:

- It is an essential component of cell membranes and nerve fibers.
- It is a primary energy source, providing up to 70% of our total energy in the resting state.
- It supports and cushions vital organs.
- All steroid hormones in the body are produced from cholesterol.
- Fat-soluble vitamins gain entry into, are stored in, and are transported through the body via fat.
- Body heat loss is minimized by the insulating effect of subcutaneous fat.

The most basic unit of fat is the fatty acid, the component used for energy production. Fatty acids occur in two forms: saturated and unsaturated. Unsaturated fats contain one (monounsaturated) or more (polyunsaturated) double bonds between carbon atoms, and

each double bond takes the place of two hydrogen atoms. A saturated fatty acid possesses no double bonds, so it has the maximum amount of hydrogen bound to the carbons. Excessive saturated fat consumption is a risk factor for numerous diseases.

Fats derived from animal sources generally contain more saturated fatty acids than fats derived from plants. Fats that are more highly saturated tend to be solids at room temperature (e.g., bacon fat, coconut oil), whereas less saturated fats tend to be liquid (e.g., olive and canola oils). And although many vegetable oils are low in saturated fats, they are often used in foods as hydrogenated shortening. The process of hydrogenation adds hydrogen to the fat, increasing its saturation.

Fat Consumption

Fat can enhance food's palatability by absorbing and retaining flavors and by affecting the food's texture. For this reason, it is quite common in our diets, and extremely low-fat diets are unpalatable. The dietary fat intake for both men and women was as high as 45% of total calories consumed in 1965 but has decreased to about 33% to 34% as reported in the most recent National Health and Nutrition Examination Survey (NHANES).³³ Most likely this decrease was attributable to recent media attention on the health risks of dietary fat. Most nutritionists recommend that fat consumption not exceed 35% of total calories consumed. *Dietary Guidelines for Americans, 2015-2020*, produced by the U.S. Departments of Health and Human Services and Agriculture, advises us to limit saturated fats to less than 10% of total caloric intake and replace with polyunsaturated and monounsaturated fats, to limit cholesterol to less than 300 mg per day, and to eliminate trans fatty acids.¹⁶ Recent research has shown that not all fats categorized by degree of saturation have the same health effect. For example, not all saturated fats are "bad" fats; the saturated fat stearic acid does not negatively affect serum cholesterol levels.^{34,35}

RESEARCH PERSPECTIVE 15.3

Low-Carb and Low-Fat Diets

Trends in dieting strategies for weight loss often become popular before the scientific community has an opportunity to appropriately test their effects not only on weight loss and maintenance but also on numerous other health outcomes as well. To date, only the *Dietary Approaches to Stop Hypertension (DASH)* diet, which is low in fat and sodium and high in whole grains, fruits, and vegetables, has been scientifically evaluated and shown to improve cardiovascular health. Low-carbohydrate diets, in which the person adhering to the diet consumes <40 g of carbohydrates per day, have become increasingly popular in recent years. However, the cardiovascular effects of this approach are relatively unknown. Prospective studies (which apply an intervention or stimulus and then follow the participants into the future to measure outcomes) of low-carbohydrate diets have produced conflicting results, and few randomized controlled studies have examined the effects of carbohydrate restriction on cardiovascular disease risk factors in a diverse population.

In 2014, a research team from Tulane University conducted a randomized, parallel group trial to examine the effects of a 12-month low-carbohydrate diet compared to a low-fat diet.⁶ In this study, 148 men and women ages 22 to 75 years with a body mass index (BMI) of 30 to 45 kg/m² from diverse socioeconomic and ethnic backgrounds were assigned to one of two diet groups: low fat or low carbohydrate. The 73 participants assigned to the low-fat diet were instructed to maintain less than 30% of their daily energy intake from fat and 55% from carbohydrates. The 75 participants assigned to the low-carbohydrate diet were instructed to eat less than 40 g of carbohydrate daily. Subjects met with a dietician weekly for the first 4 weeks, followed by group counseling sessions every other week for the next 5 weeks, and then monthly for the last 6 months of the study. The study team collected data on weight, cardiovascular disease risk factors, and dietary composition at baseline, and 3, 6, and 12 months into the intervention. By the end of the yearlong intervention, 60 participants in the low-fat group and 59 in the low-carbohydrate group had completed the entire year.

After 12 months, participants in the low-carbohydrate group had greater decreases in weight, fat mass, and low-density lipoprotein cholesterol and greater increases in high-density lipoprotein cholesterol compared to the low-fat diet group. These data indicate that low-carbohydrate diets may be more effective for weight loss and reducing cardiovascular disease risk factors compared to a low-fat diet, even in a very large and diverse group of participants. The study concluded that restricting carbohydrates may be a good option for people seeking to lose weight and reduce cardiovascular disease risk.

Fat Intake and Performance

For the athlete, fat is especially important as an energy source. Muscle and liver glycogen stores in the body are limited, so the use of fat (or FFA) for energy production can delay exhaustion. Clearly, any change that allows the body to use more fat would be an advantage, particularly for endurance performance. In fact, one adaptation that occurs in response to endurance training is an increased ability to use fat as an energy source. Unfortunately, merely eating fat does not stimulate the muscles to burn fat. Instead, eating fatty foods tends only to elevate plasma triglycerides, which then must be broken down before the FFA can be used for energy production. To increase the use of fat, the FFA levels in the blood, not the triglyceride levels, must be increased.

Highly trained athletes can adapt to a high-fat diet. But is this beneficial to overall performance? We have just seen how glycogen loading can improve endurance performance. Does fat loading have similar benefits? While several studies have demonstrated limited benefit of a high-fat versus a high-carbohydrate diet, most studies have shown either no benefit or decreased performance. The body adapts to a high-fat diet by increasing the supply of fat and the capacity for fat oxidation in the muscle, thus increasing fat oxidation during exercise. But this usually occurs at the expense of a reduction in muscle glycogen stores, thus negating any beneficial effects. Drawing conclusions from the studies conducted so far is complicated by the fact that the type of fat (medium-chain versus long-chain triglycerides) and the duration of the high-fat diet intervention (less than a week versus several weeks or more) have varied widely in these studies.

Protein

Protein is a class of nitrogen-containing compounds formed by amino acids. Protein serves numerous functions in our bodies:

- It is the major structural component of the cell.
- It is used for growth, repair, and maintenance of body tissues.
- Hemoglobin, enzymes, and many hormones are produced from protein.
- It is one of the three primary buffers in the control of acid–base balance.

- Proteins in the plasma help maintain normal blood osmotic pressure.
- Antibodies for disease protection are formed from protein.
- Energy can be produced from protein.

Twenty amino acids have been identified as necessary for human growth and metabolism (table 15.2). Of these, 11 (for children) or 12 (for adults) are termed **nonessential amino acids**, meaning that our bodies synthesize them, so we don't rely on dietary intake for their supply. The remaining eight or nine are termed **essential amino acids** because we cannot synthesize them; thus, they are an essential part of our daily diets. Absence of one of these essential amino acids from the diet precludes formation of any proteins that contain that amino acid, and thus any tissue requiring those proteins cannot be maintained.

TABLE 15.2 Essential and Nonessential Amino Acids

Essential	Nonessential
Isoleucine	Alanine
Leucine	Arginine
Lysine	Asparagine
Methionine	Aspartic acid
Phenylalanine	Cysteine
Threonine	Glutamic acid
Tryptophan	Glutamine
Valine	Glycine
Histidine (children) ^a	Proline
	Serine
	Tyrosine
	Histidine (adult) ^a

^aHistidine is not synthesized in infants and young children, so it is an essential amino acid for children but not for adults.

A dietary protein source that contains all the essential amino acids is called a complete protein. Meat, fish, poultry, eggs, and milk are examples. The proteins in vegetables and grains are called incomplete proteins because they do not supply all the essential amino acids. This concept is important for people on vegetarian diets (discussed later in this chapter). However, combining several incomplete protein sources at one meal should resolve this problem.

Protein Consumption

Protein accounts for approximately 15% of the total calories consumed per day in the United States. The Recommended Dietary Allowances (RDAs) for protein are 0.95 g/kg of body weight per day

for 4- to 13-year-olds, 0.85 g/kg per day for 14- to 18-year-olds, and 0.80 g/kg per day for adults. Men typically require more protein than women because men generally weigh more and have greater muscle mass. However, men generally eat more food per day to sustain their weight and muscle mass. Thus, an allowance of 0.8 g/kg of body weight is considered appropriate for both men and women.

Protein Intake and Performance

Should athletes who are training for muscular strength and endurance and aerobic endurance increase their protein intake? Amino acids are the body's building blocks, so protein is essential for the growth and development of body tissues. Over the years, nutritionists and physiologists argued against the need for supplementing proteins for optimal sport performance. It was generally believed that the RDA of 0.8 g of protein per kilogram of body weight each day would adequately meet the demands of hard training. However, in sports that rely on the maintenance or building of muscle mass, the RDA is probably inadequate.

Studies using metabolic-tracer and nitrogen-balance technologies have shown that the overall protein and specific amino acid requirements are higher for individuals in training than for normally active people. The role of protein differs for endurance- and strength-trained athletes. It appears that strength-training individuals need up to 2.1 times the RDA, or about 1.6 to 1.7 g of protein per kilogram of body weight per day, whereas athletes engaging in endurance training need 1.2 to 1.4 g of protein per kilogram of body weight per day.² Most athletes cross-train (endurance athletes do some strength training and power athletes perform some aerobic exercise); therefore, a general recommendation of 1.2 to 1.7 g of protein per kilogram of body weight per day has been recommended,²⁷ which is close to the 1.4 to 1.6 g/kg per day supported by empirical studies. While endurance exercise places greater demand on protein to increase mitochondrial content and as an auxiliary fuel, strength training requires additional amino acids as the building blocks for muscle protein synthesis. Of course, there could be exceptions to this for athletes who are just starting a new, rigorous training program or who are doing very intense, long-duration workouts. Protein timing is important for athletes, and current advice is to consume about 20 g of